

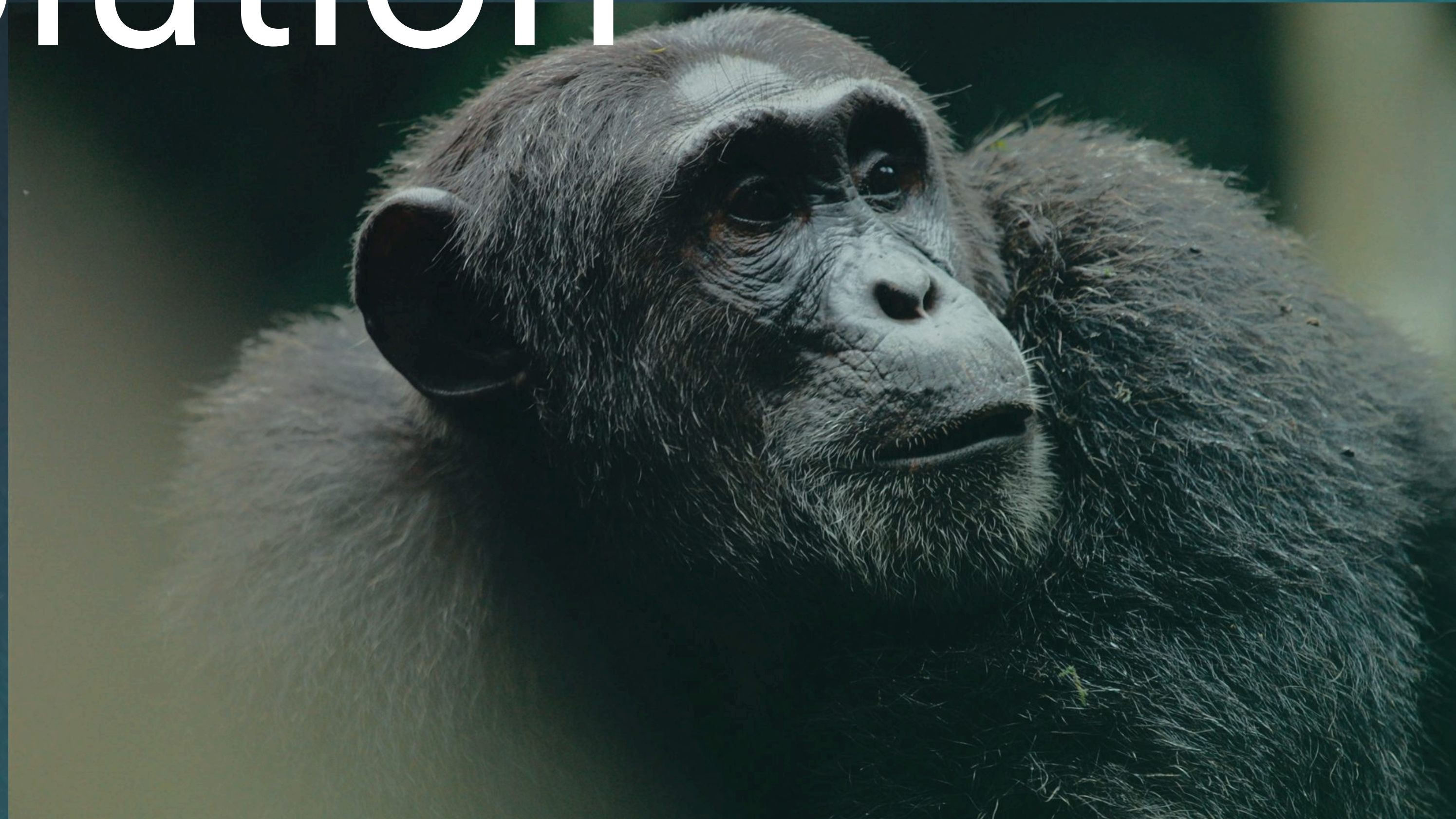
From Neural Networks to Digital Brains

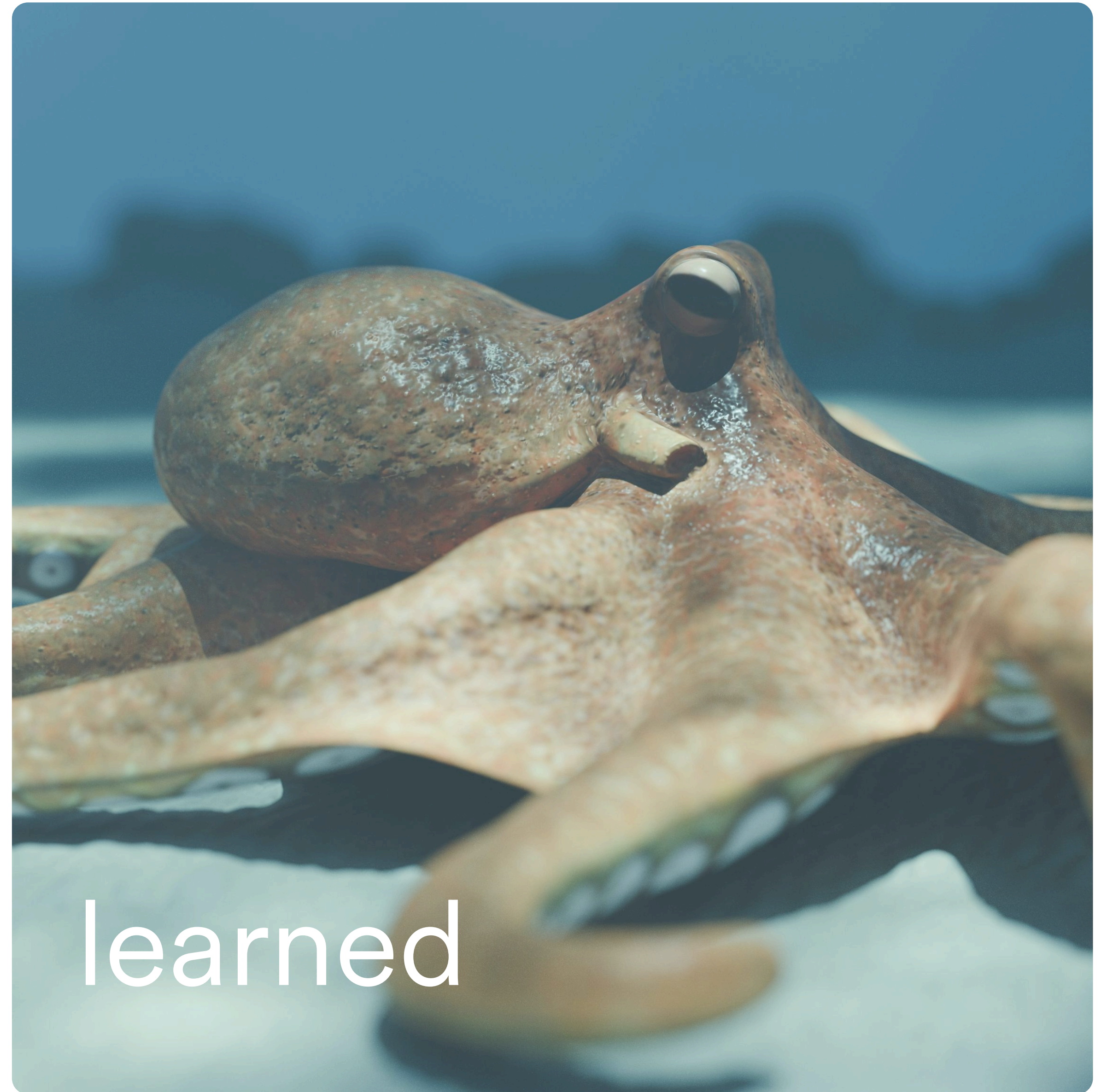
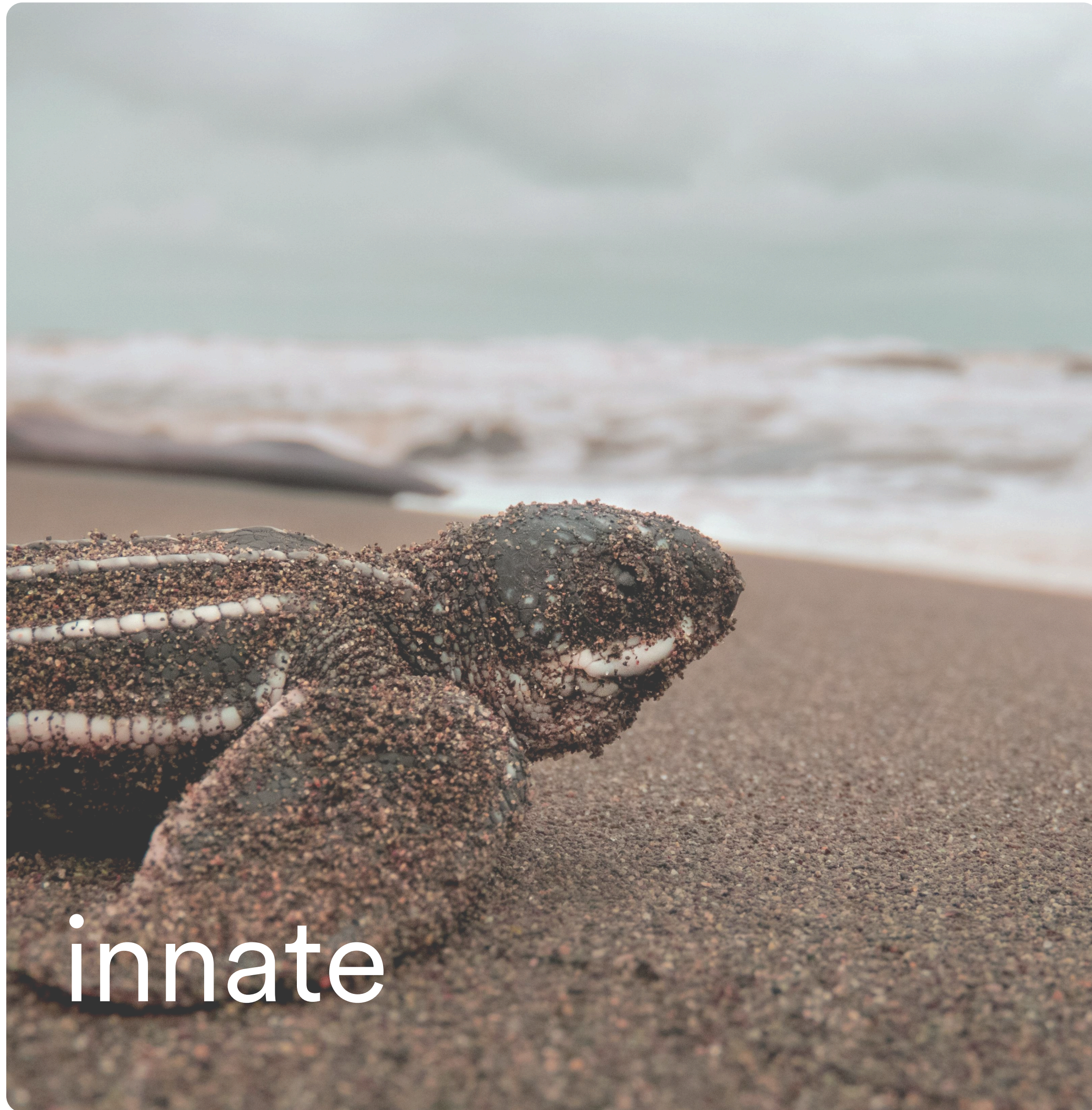


A macro photograph of three ants on a dark, textured surface. They are carrying a large, bright green, serrated leaf fragment. The ant in the center is positioned directly under the leaf, while the others are on either side, supporting it. The background is dark and out of focus.

How do we bridge that gap?

Evolution





PhD Thesis

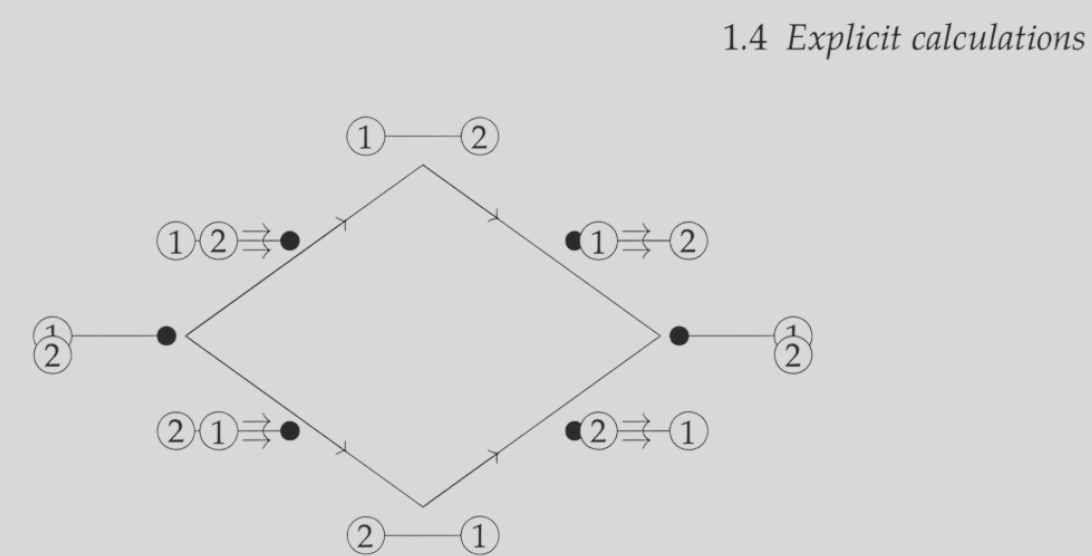
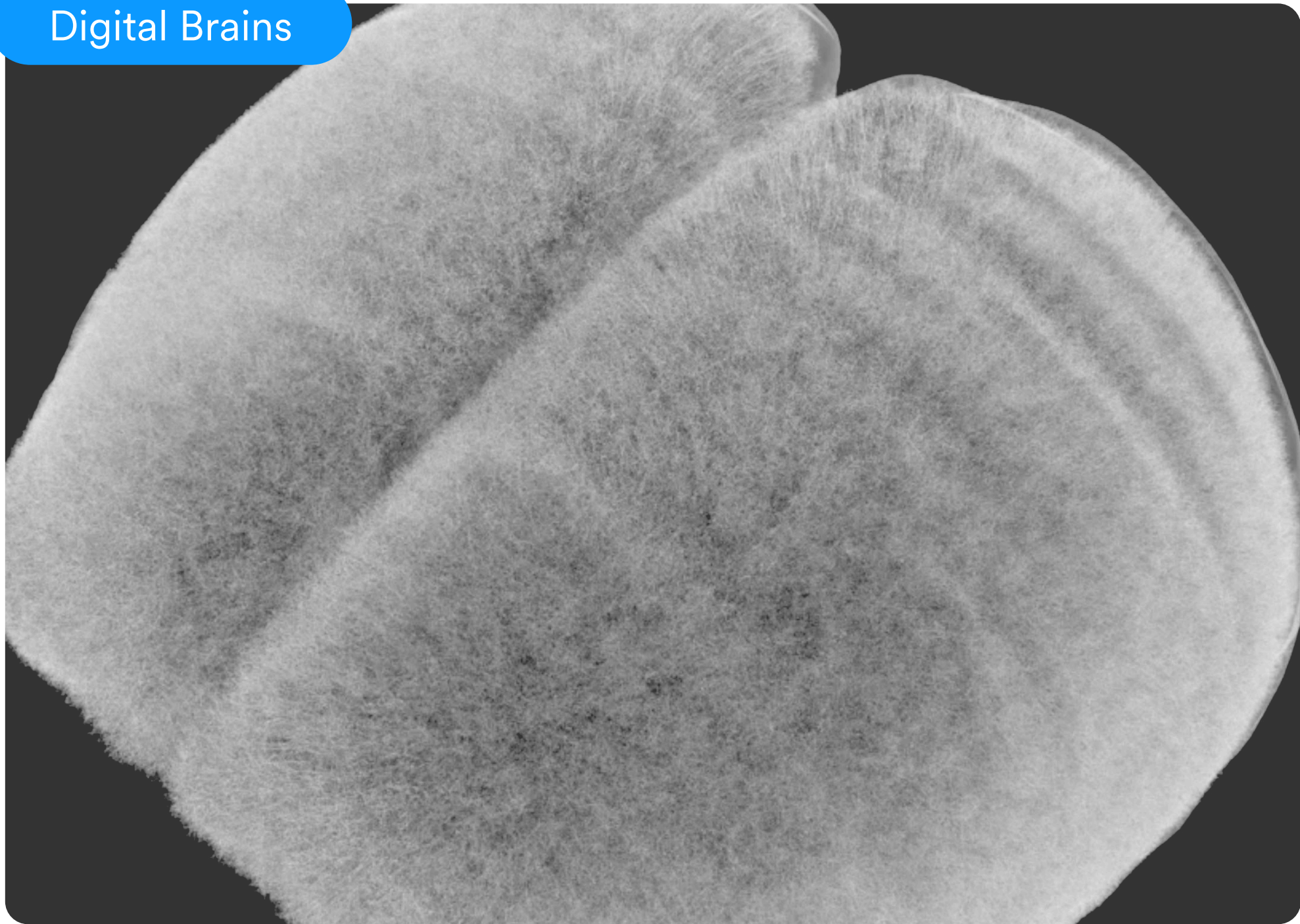


Figure 1.4: The combinatorial model of $\text{Conf}_2(I, \{0, 1\})$ consists of four edges.

Berlin 2010/17



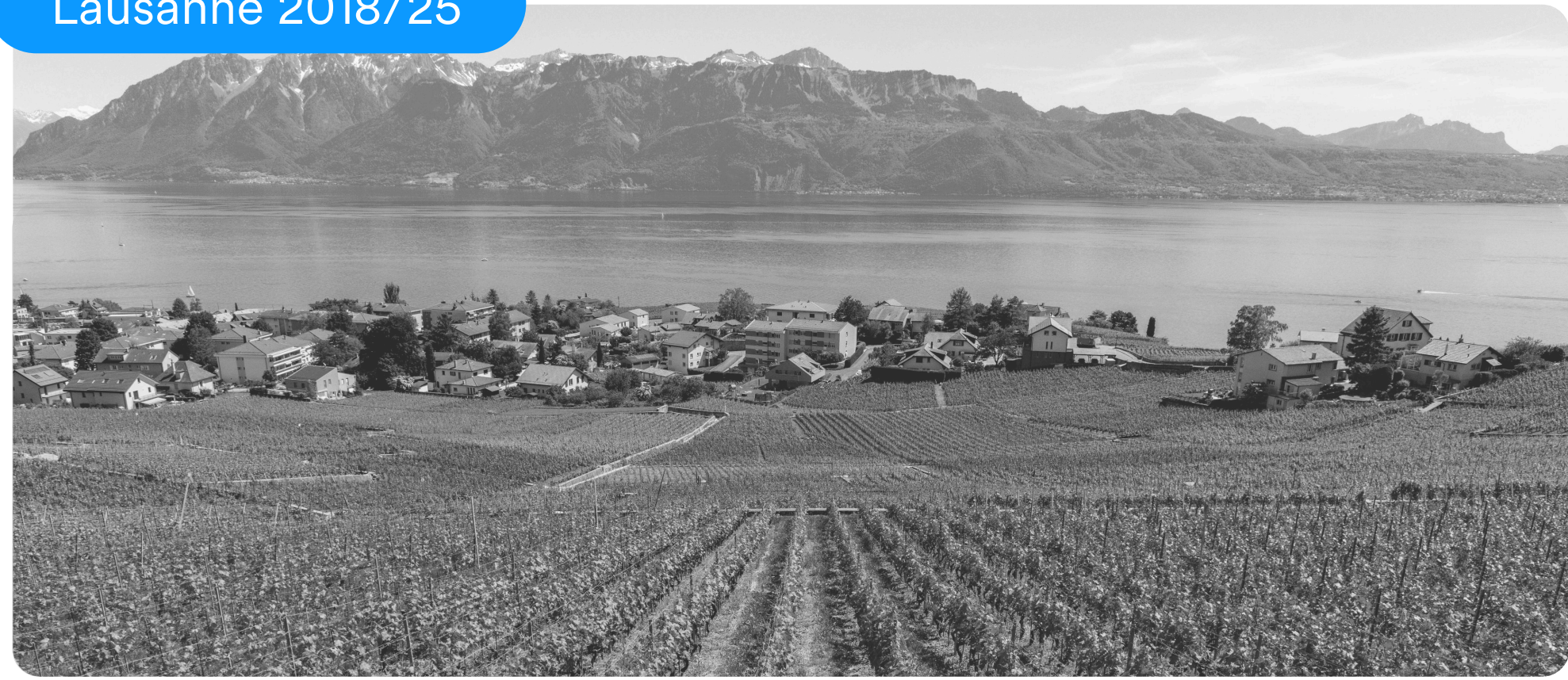
Digital Brains



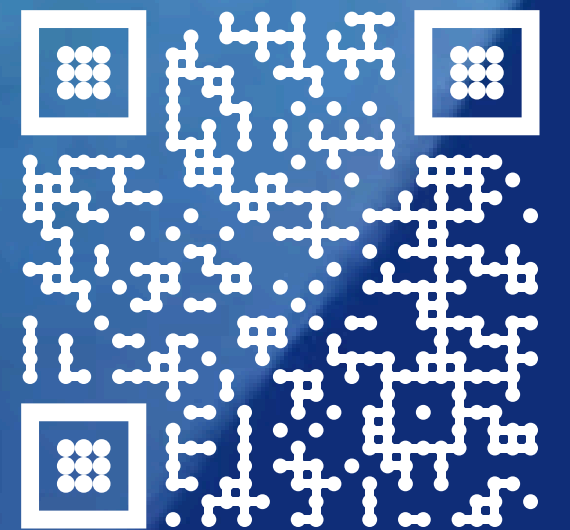
inait



Lausanne 2018/25



Digital brains

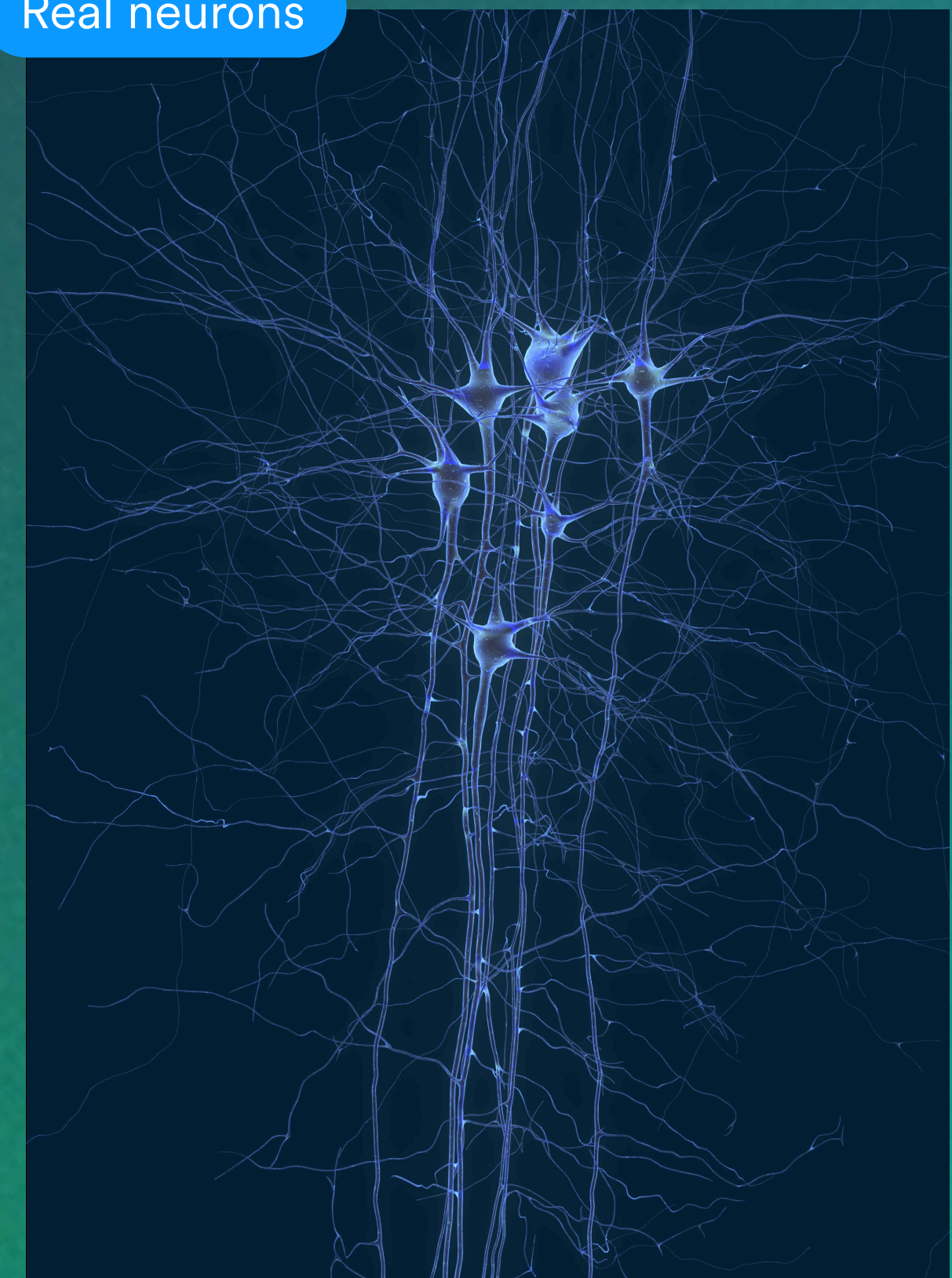


What does a neuron look like?

Deep learning neuron

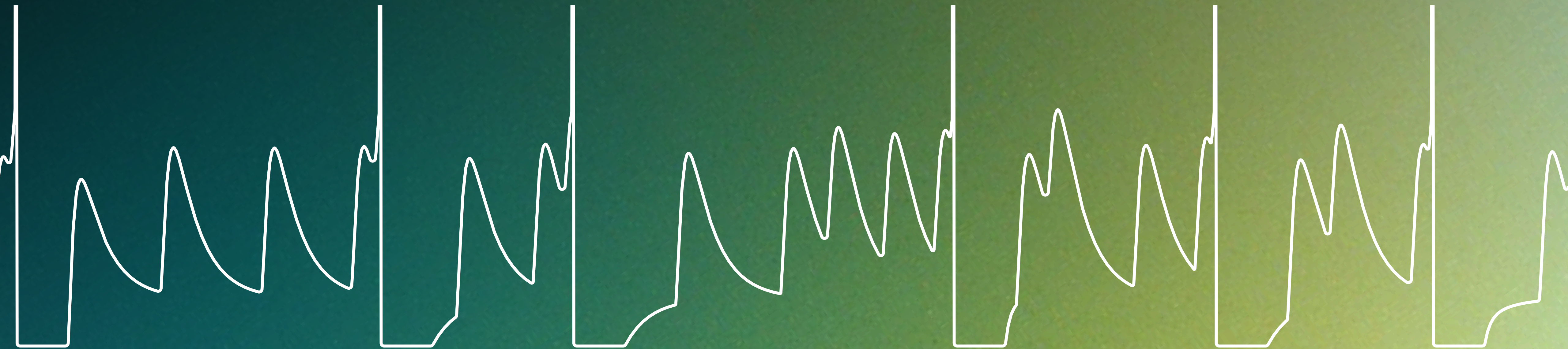


Real neurons



How does a neuron compute?

How does a neuron compute?



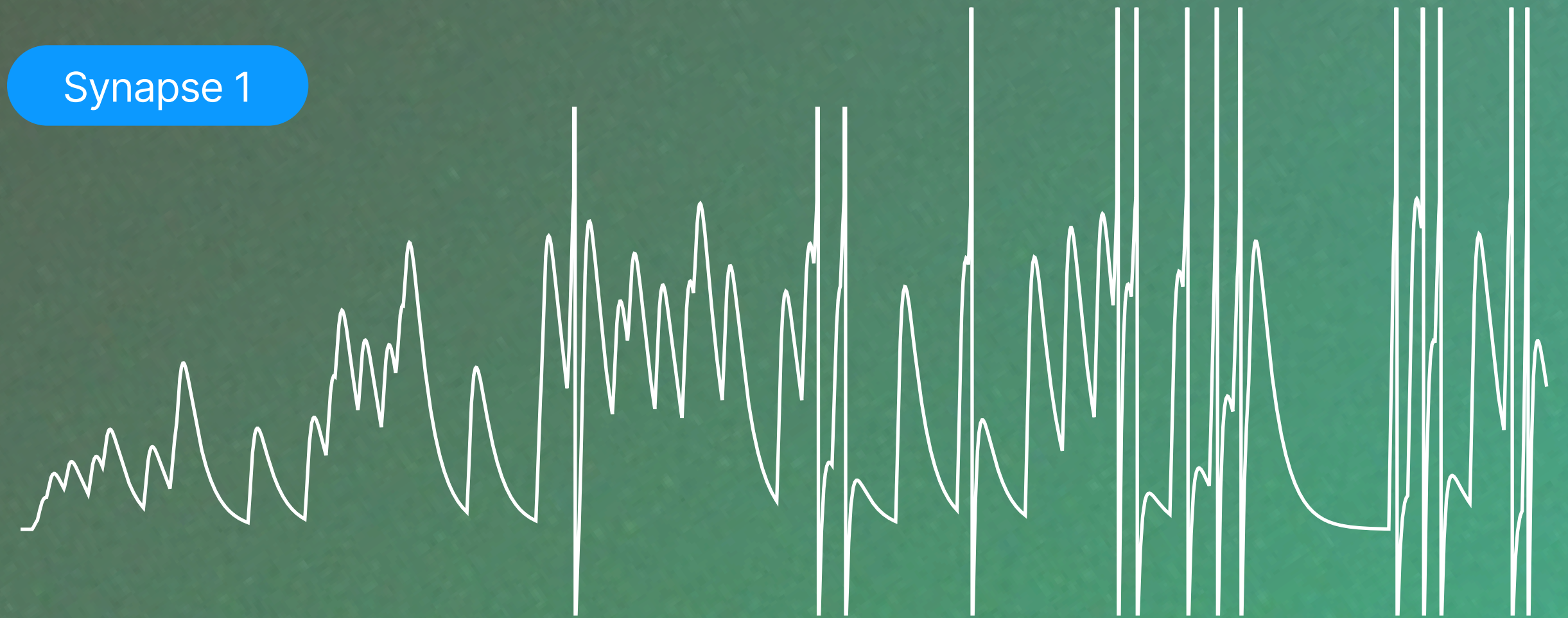
How about synapses?

Synapse input

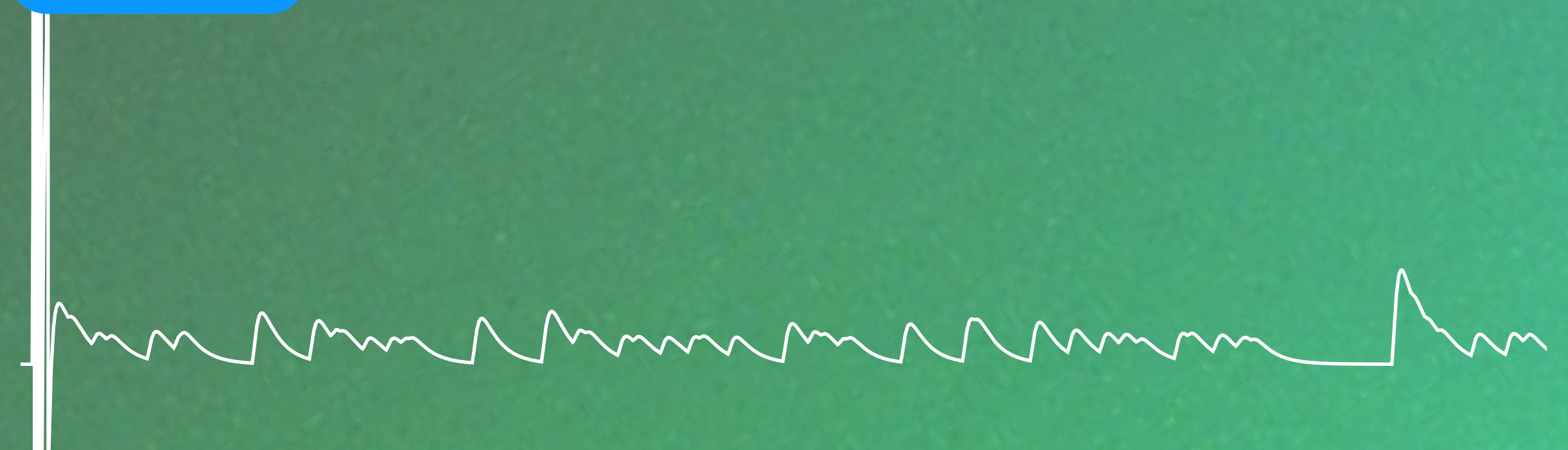


Synapses with the same weights
can behave very differently

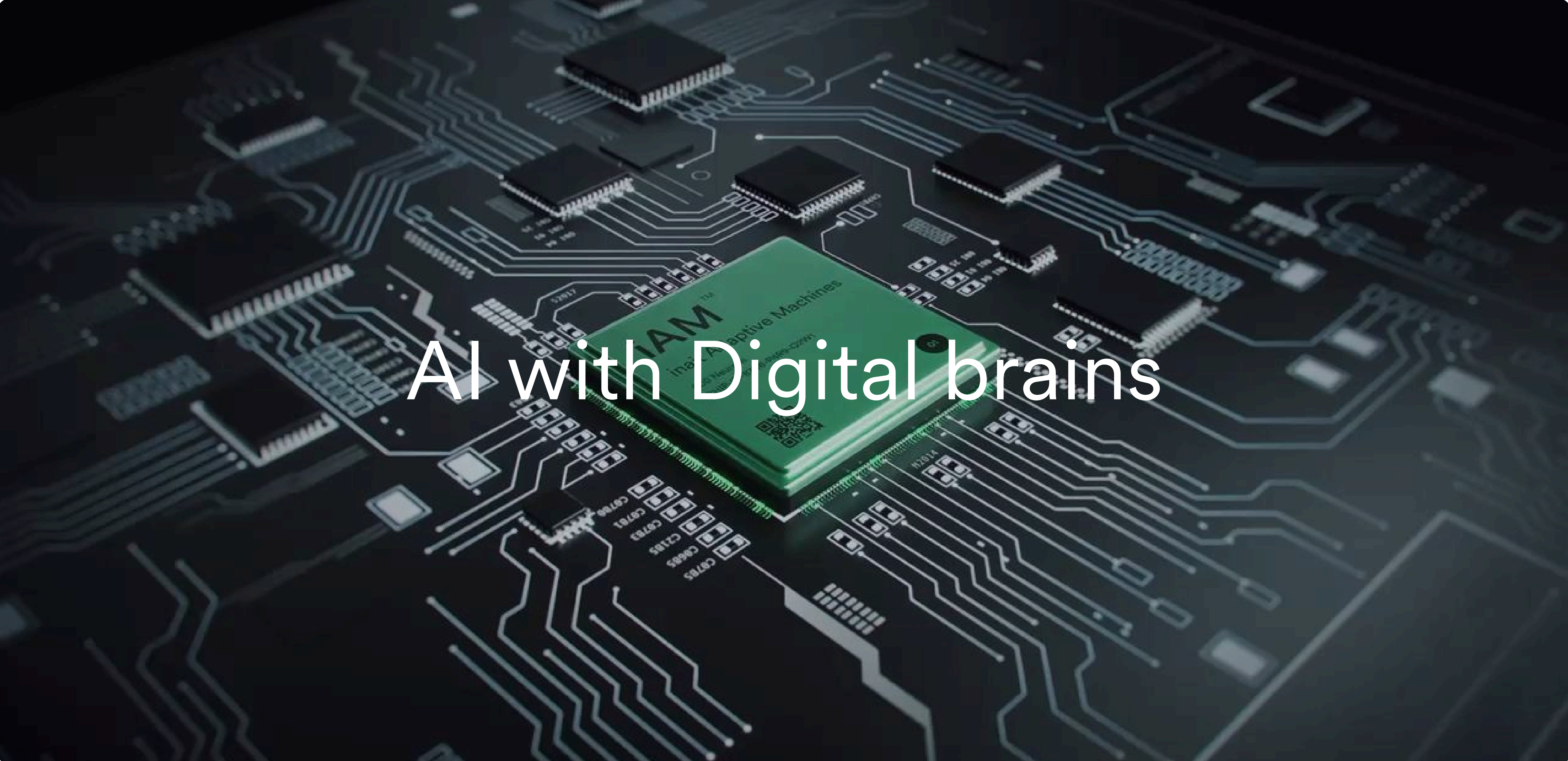
Synapse 1



Synapse 2

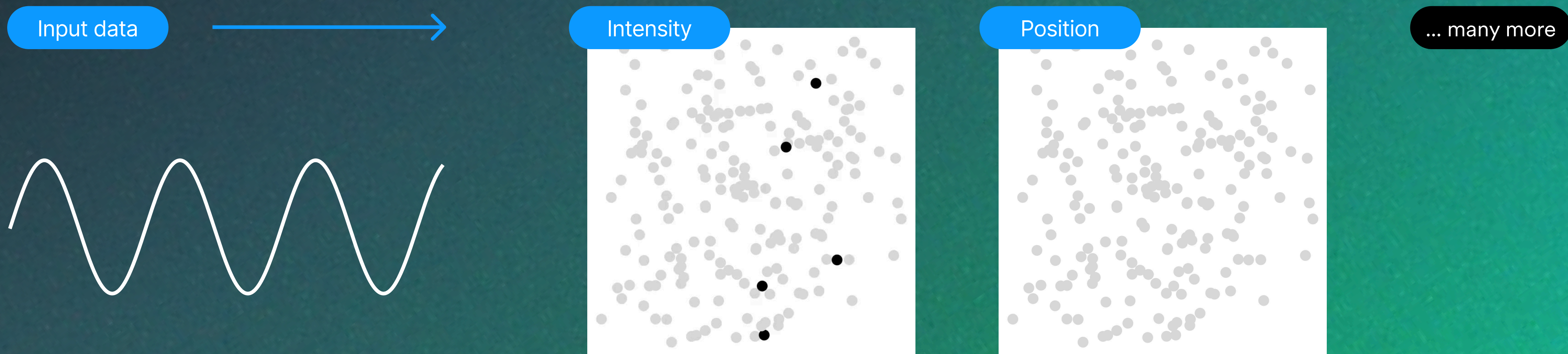


What does it look like in a network?

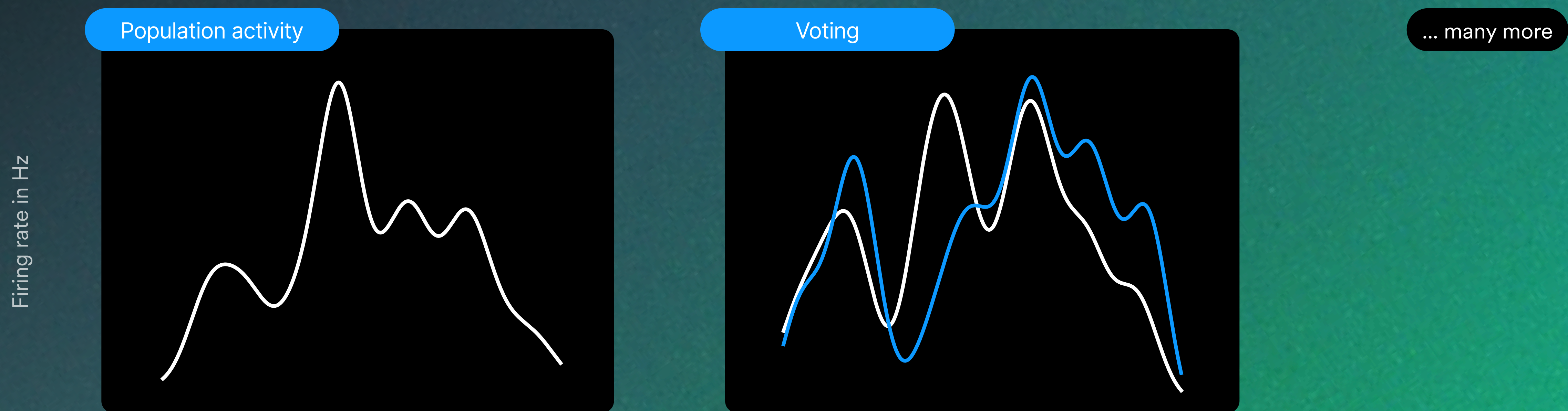


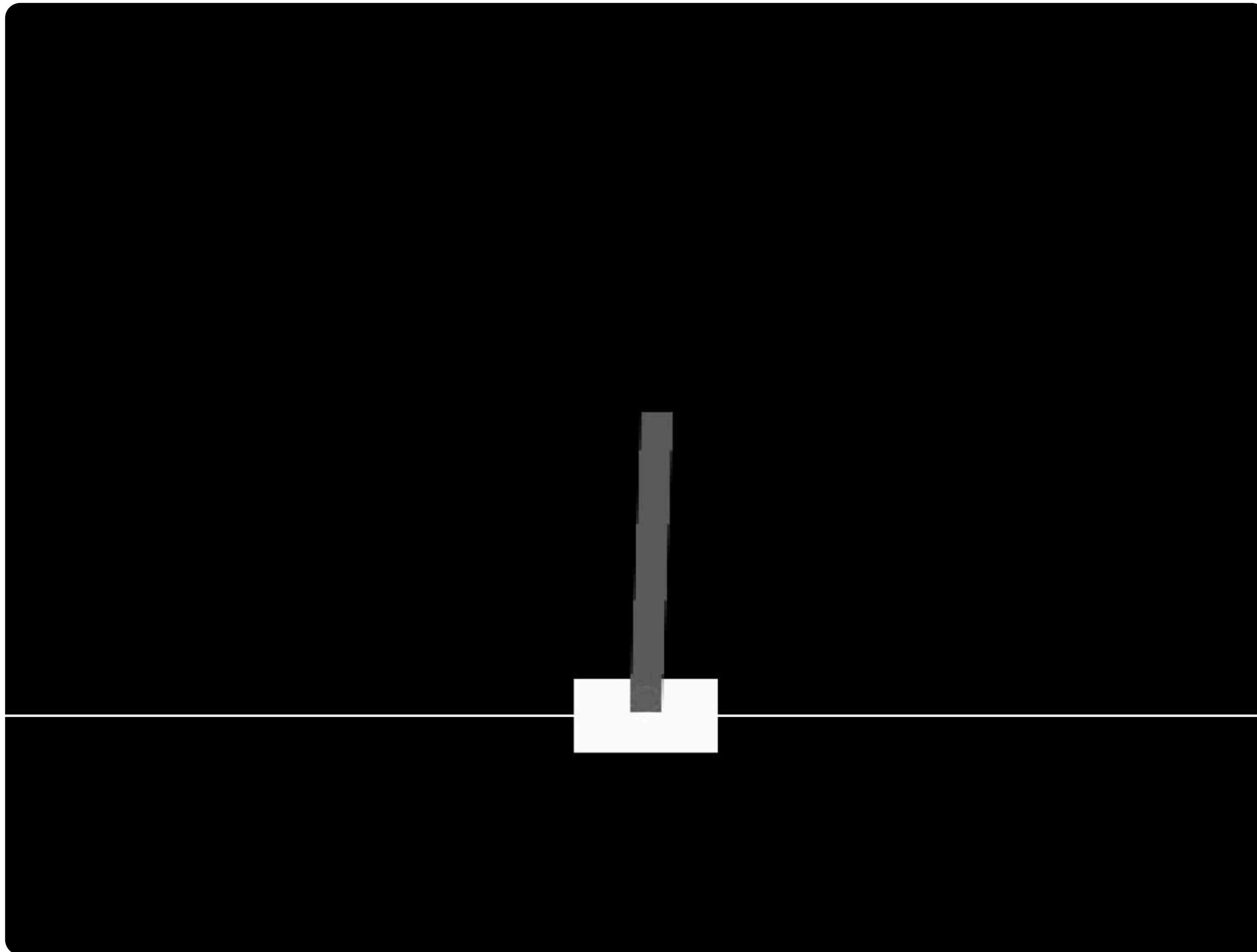
AI with Digital brains

How do you turn data into brain activity?

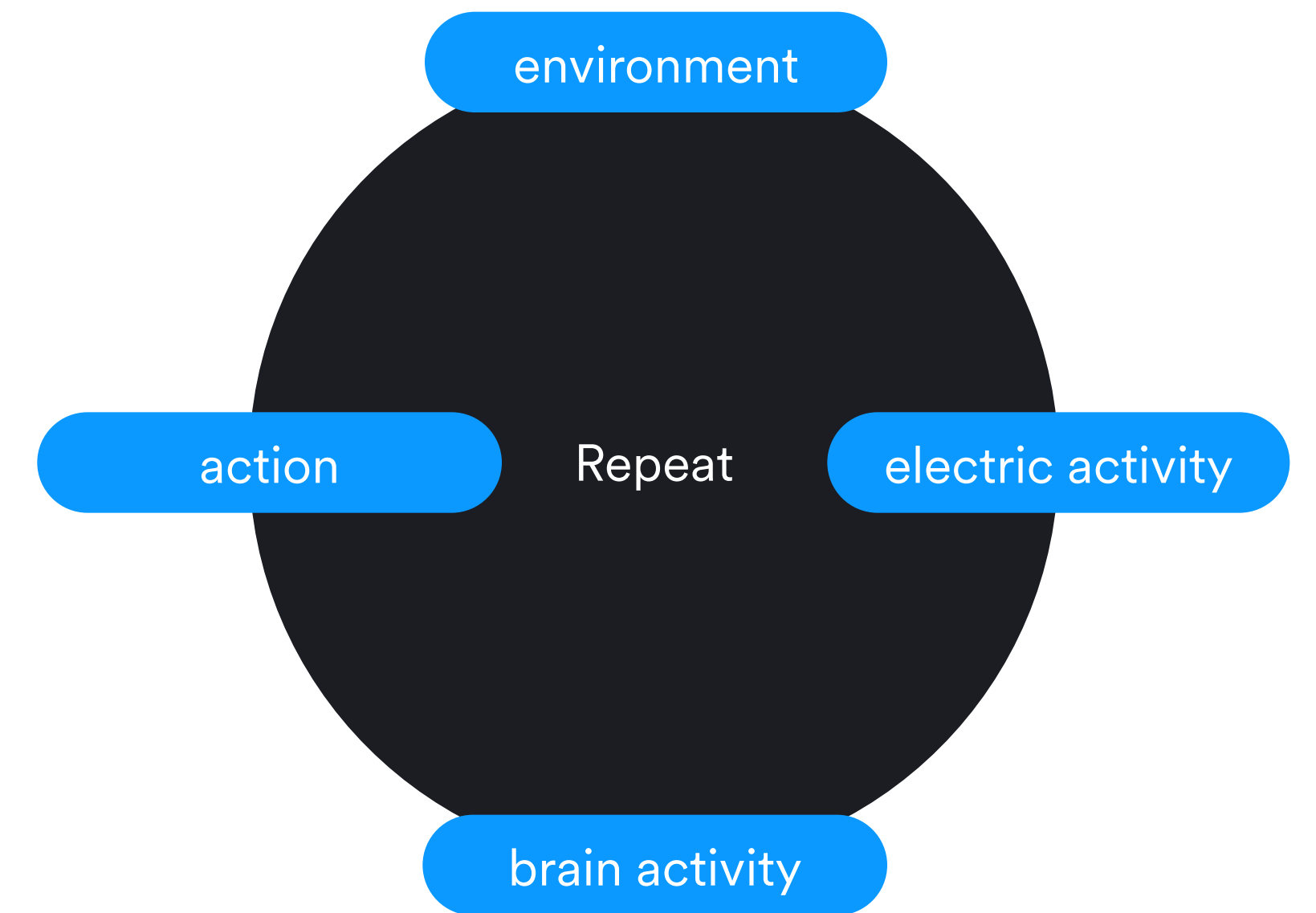


How do you turn brain activity into action?





Putting it all together



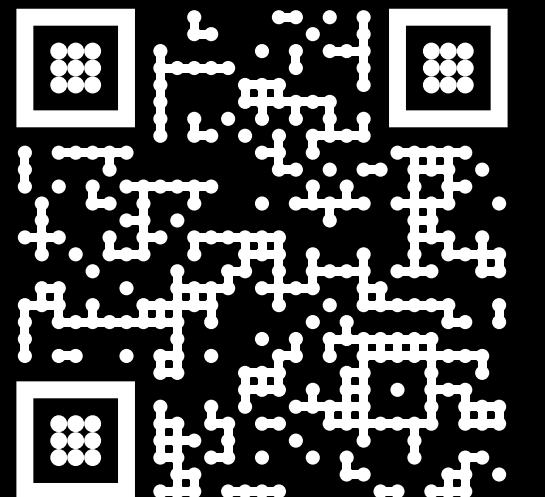
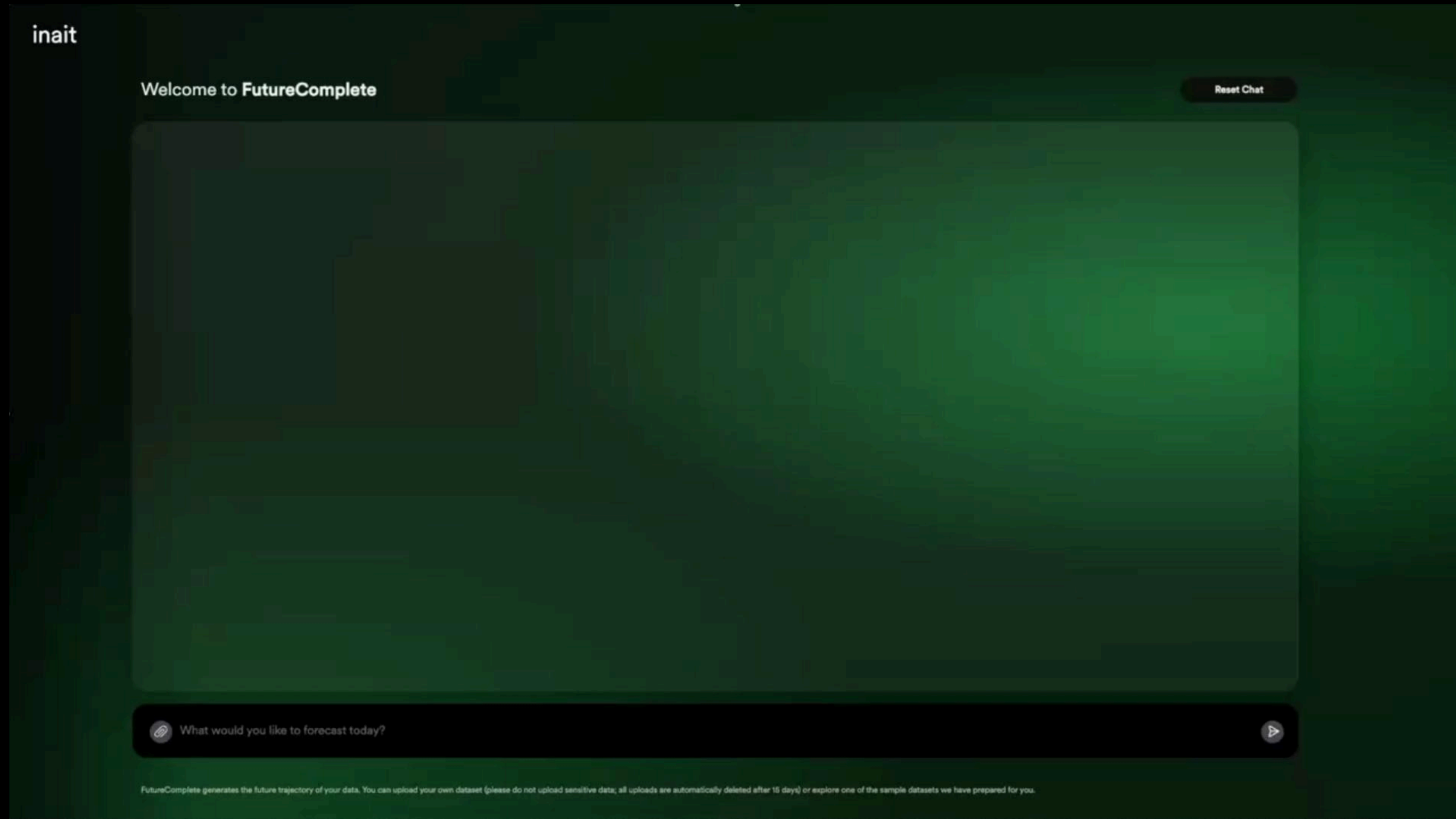


Learning saves the day!

Biological Learning

AI with Digital Brains Applications

Generation I - Forecasting



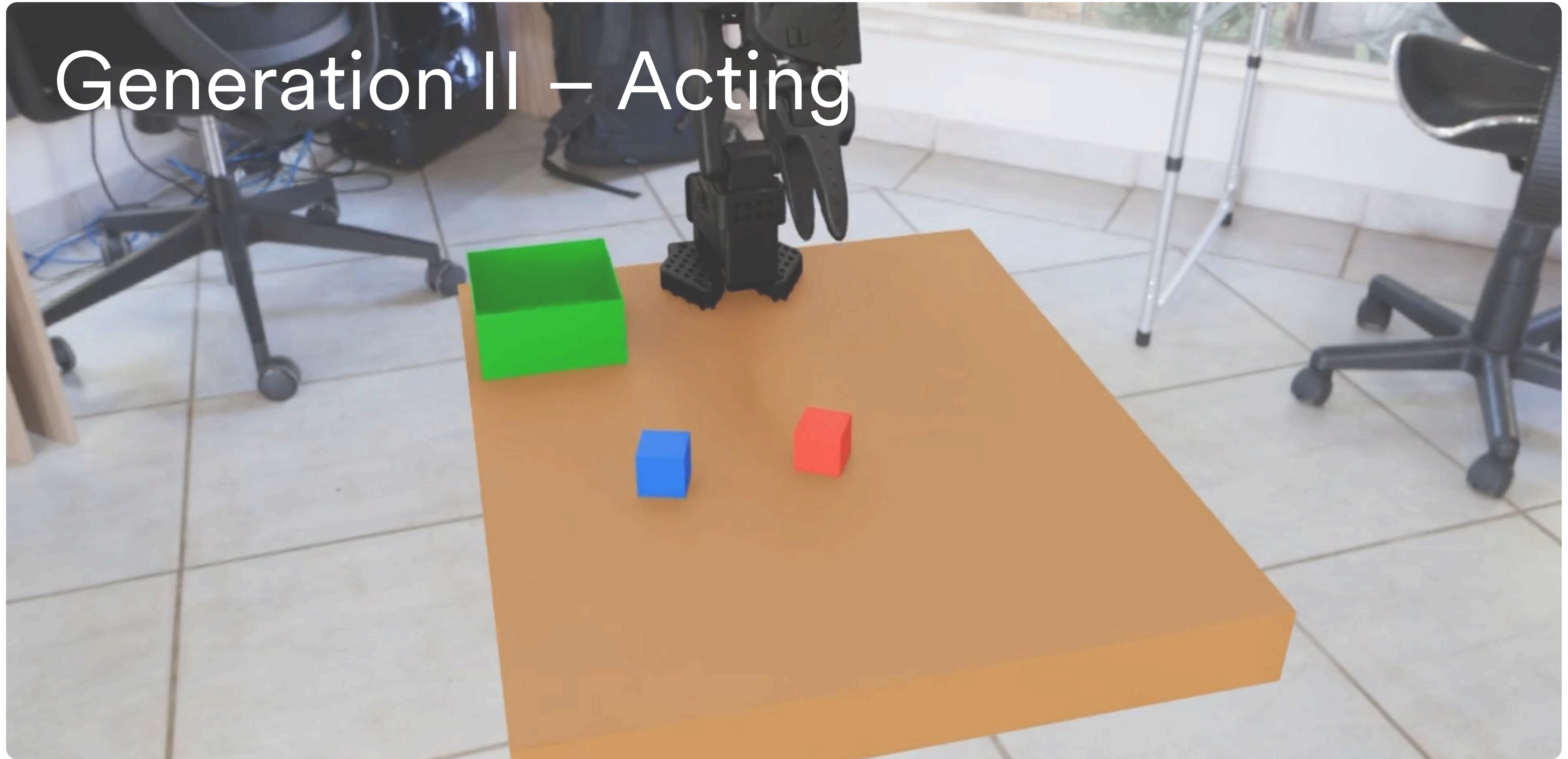
Generation II – Acting



Generation II – Acting

Game 1 – First experience of the game

Generation II – Acting





The Future

Take-aways

Complex real-time interaction requires rapid learning, from just a handful of data points

Current AI + intelligence of even tiny animals = a leap for intelligent machines

AI with Digital Brains outperforms deep learning in dynamic environments, and will power applications from complex process automation to robotics in the years to come

How would you build software for an AI that learns like a brain?

Q&A

Contact

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Follow me on LinkedIn
for more on Digital Brains



Time for lunch